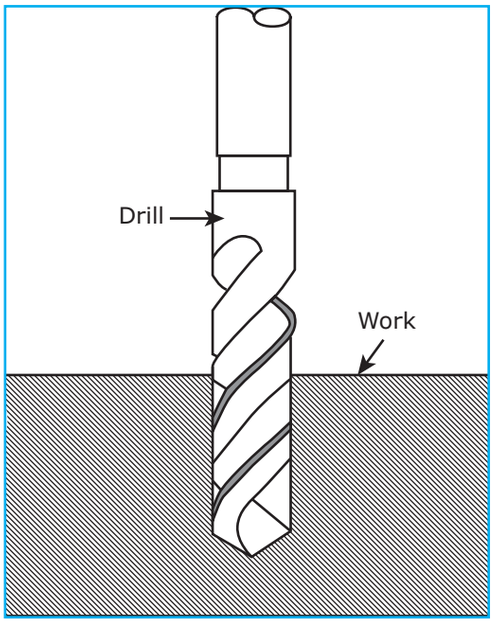
Before making a new hole by using on the work piece, first draw two lines at right angle to each other. Then a centre punch is used to make a centre on the work piece accurately.

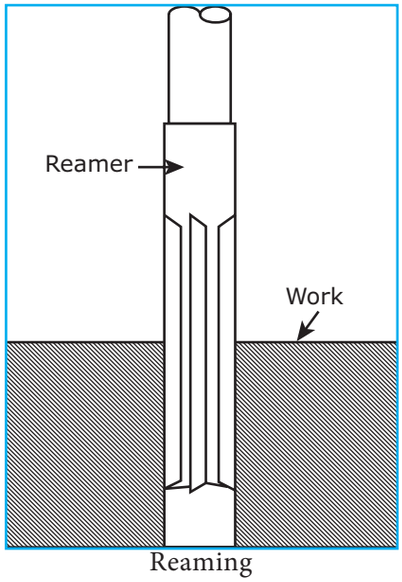
The figure shows the drilling operation



Drilling

Reaming

The size of hole made by drilling may not be accurate and the internal surface may not smooth. Reaming is an accurate way of sizing and finishing a hole which has been previously drilled by a multi – point cutting tool known as reamer. The surface obtained by reaming will be smoother and the size accurate. The speed of the spindle is made half that of drilling. Reaming removes very small amount of metal (approximately 0.375mm). In order to finish a hole and bring it to the accurate size, the hole is drilled slightly undersize. The figure shows reaming operation.



Reaming

Boring

Boring is the operation enlarging the diameter of the hole previously made. It is done for the following reasons.

To enlarge a hole by means of an adjustable cutting tool. This is done when a suitable sized drill is not available (or) the hole diameter is so large that is cannot be ordinarily drilled.

1. To finish a hole accurately and bring it to the required size.
2. To machine the internal surface of the hole already produced in casting.
3. To correct out of roundness of the hole.
4. To correct the location of the hole as the boring tool follows independent path with respect to the hole.

Boring tool is a tool with only one cutting edge. The tool is held in a boring bar which has a taper shank to fit into the spindle (or) a socket. For perfectly finishing a hole, the job is drilled undersize slightly. Boring operation in some precise drilling machine is performed to enlarge the holes to an accuracy of 0.00125mm. The spindle speed during the boring operation should be adjusted to be lesser than that of reaming.

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